

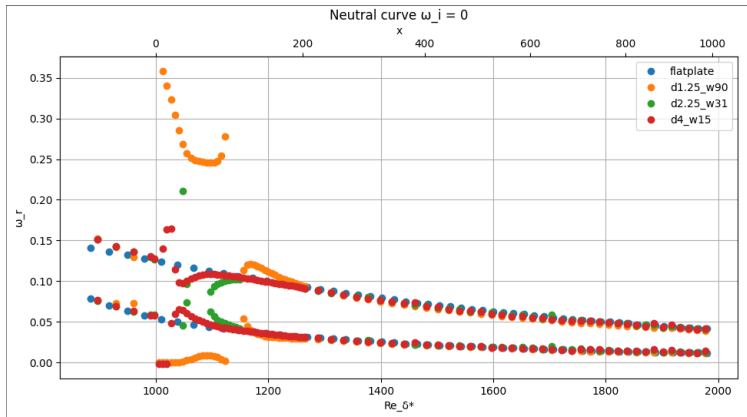
IMPERIAL

Neutral stability gaps

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Neutral stability curves

- orr sommerfeld analysis with parallel flow assumption (e.g. no weakly non parallel effects).
- run for cases flat plate, $d, w \in \{(1.25, 90), (2.25, 31), (4, 15)\}$
- there are a few points where the flow is locally convectively stable (small decay after the gap).



Neutral stability curves

- my feeling to understand the downstream evolution of some of the n-factor curves is that just after the gap, many frequencies have grown a lot (observing the neutral stability curve), and when we recover the blasius-like profile in the downstream BL, only a few frequencies will still be able to grow while many others will decay, resulting in a “compensation” of amplitudes.

